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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 8: Equilibria

Lesson	Title (MINESEC progression)
1	Reversible reactions and dynamic equilibrium
2	The position of equilibrium and Le Chatelier's principle
3	Factors affecting the position of equilibrium
4	The equilibrium law
5	Calculating Kc and equilibrium concentrations
6	Kp and partial pressures
7	Significance of K and factors affecting it
8	Heterogeneous equilibria
9	Temperature, enthalpy change and K
10	Redox systems: half-cells and cells
11	Redox equilibria and balancing redox equations
12	Types of half-cells
13	Standard electrode potentials and the SHE
14	Other reference electrodes
15	Electrochemical cells: the Daniell cell
16	Cell diagrams and cell e.m.f.
17	Uses of standard electrode potentials
18	Factors affecting electrode potentials
19	Theories of acids and bases
20	pH and its measurement
21	The ionic product of water
22	Strong acids and bases
23	Weak acids and bases
24	pH of mixtures of acids and bases
25	Acid–base indicators
26	Titration curves (one equivalence point)
27	Titration curves with two equivalence points
28	Buffer solutions
29	pH of acidic buffers
30	pH of alkaline buffers
31	Salt hydrolysis
32	Solubility and solubility product
33	The common ion effect

Lesson 1: Reversible reactions and dynamic equilibrium

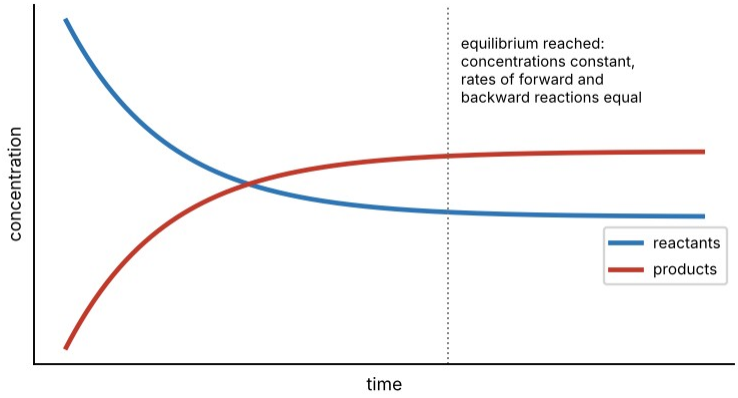
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define dynamic equilibrium and state its features

Lesson notes

- A **reversible reaction** can go in both directions: $N_2 + 3H_2 \rightleftharpoons 2NH_3$; $CuSO_4 \cdot 5H_2O \rightleftharpoons CuSO_4 + 5H_2O$ (blue ⇌ white on heating/adding water).
- **Static equilibrium:** no movement at all. **Dynamic equilibrium:** forward and reverse reactions continue at equal rates, so concentrations stay constant.



KEY POINTS

Conditions/features of dynamic equilibrium: a closed system; constant temperature; rate forward = rate backward; macroscopic properties (colour, pressure, concentration) constant; can be reached from either side

Worked examples

Example: Explain why a bottle of fizzy drink stays fizzy while sealed but goes flat when opened.

Sealed: $CO_2(g) \rightleftharpoons CO_2(aq)$ is at dynamic equilibrium (closed system). Open: CO_2 escapes, so equilibrium is never reached and more CO_2 leaves the solution

Practice Questions: Lesson 1

1. State three features of a system in dynamic equilibrium.

Answers: *1. closed system; forward and reverse rates equal; concentrations constant; can be reached from either side*

Lesson 2: The position of equilibrium and Le Chatelier's principle

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ state Le Chatelier's principle and use it to predict shifts

Lesson notes

- The **position of equilibrium** describes the relative amounts of reactants and products at equilibrium: "lies to the right" means mostly products.

KEY POINTS

Le Chatelier's principle: if a system at equilibrium is subjected to a change in conditions, the position of equilibrium shifts so as to oppose (reduce the effect of) the change

Worked examples

Example: For $2CrO_4^{2-}$ (yellow) + $2H^+ \rightleftharpoons Cr_2O_7^{2-}$ (orange) + H_2O, predict the effect of adding acid, then alkali.

Adding H^+: equilibrium shifts right to remove H^+: solution turns orange

Adding OH^-: removes H^+, equilibrium shifts left: solution turns yellow

Practice Questions: Lesson 2

1. For $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$, predict the colour change when NaOH is added.

Answers: *1. H^+ is removed, equilibrium shifts left: orange → yellow*

Lesson 3: Factors affecting the position of equilibrium

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ predict the effects of changing conditions on equilibrium position

Lesson notes

Change	Effect on position	Effect on K
Increase concentration of a reactant	shifts right (to use it up)	none
Increase pressure (gases)	shifts to the side with fewer gas molecules	none
Increase temperature	shifts in the endothermic direction	changes
Add a catalyst	no change; equilibrium reached faster	none

Worked examples

Example: For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92$ kJ mol^{-1}, explain the conditions of the Haber process (450 °C, 200 atm, iron catalyst).

High pressure: 4 gas molecules → 2, so the yield increases (but very high pressure is costly and dangerous)

Low temperature favours the exothermic forward reaction (higher yield) but is too slow; 450 °C is a compromise between yield and rate

Iron catalyst speeds up the attainment of equilibrium; the unreacted gases are recycled

- Similar reasoning applies to the **Contact process** ($2SO_2 + O_2 \rightleftharpoons 2SO_3$, $\Delta H = -197$ kJ mol^{-1}: 450 °C, 1–2 atm, V_2O_5 catalyst).

Practice Questions: Lesson 3

1. For $N_2O_4 \rightleftharpoons 2NO_2$ ($\Delta H = +58$ kJ mol^{-1}), state the effect of increasing temperature and of increasing pressure.

Answers: *1. temperature: shifts right (more NO_2, darker); pressure: shifts left (fewer gas molecules)*

Lesson 4: The equilibrium law

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ write Kc expressions and deduce their units

Lesson notes

KEY POINTS

For aA + bB ⇌ cC + dD at equilibrium: $K_c = [C]^c[D]^d / ([A]^a[B]^b)$, where [ ] = equilibrium concentration in mol dm^{-3}. Kc is constant at a given temperature

Worked examples

Example: Write Kc expressions for (a) $H_2 + I_2 \rightleftharpoons 2HI$ (b) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ and give the units.

(a) $K_c = [HI]^2/([H_2][I_2])$; no units

(b) $K_c = [NH_3]^2/([N_2][H_2]^3)$; units $(\text{mol dm}^{-3})^{-2} = \text{mol}^{-2}\ \text{dm}^6$

Practice Questions: Lesson 4

1. Write Kc and its units for $2SO_2 + O_2 \rightleftharpoons 2SO_3$.

Answers: *1. $[SO_3]^2/([SO_2]^2[O_2])$; $\text{mol}^{-1}\ \text{dm}^3$*

Lesson 5: Calculating Kc and equilibrium concentrations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ calculate Kc and equilibrium concentrations

Lesson notes

EXAM TIP

Use an "ICE" table: Initial moles, Change (from the equation's ratio), Equilibrium moles; then divide by volume.

Worked examples

Example: 1.00 mol of ethanoic acid and 1.00 mol of ethanol were mixed; at equilibrium, 0.667 mol of ethyl ethanoate was present. Find Kc for $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$.

Equilibrium: acid 0.333, alcohol 0.333, ester 0.667, water 0.667 mol (volume cancels)

Kc = (0.667 × 0.667)/(0.333 × 0.333) = 4.0

Example: For $H_2 + I_2 \rightleftharpoons 2HI$, Kc = 49 at 450 °C. 1.0 mol of H_2 and 1.0 mol of I_2 are heated in a 1.0 dm^3 flask. Find the equilibrium amounts.

Let x mol H_2 react: $H_2 = I_2 = 1 - x$, $HI = 2x$

$(2x)^2/(1 - x)^2 = 49 \rightarrow 2x/(1 - x) = 7 \rightarrow x = 7/9 = 0.778$

$H_2 = I_2 = 0.222$ mol; HI = 1.56 mol

Practice Questions: Lesson 5

1. 2.0 mol H_2 and 2.0 mol I_2 in 1.0 dm^3 form 3.0 mol HI at equilibrium. Find Kc.

Answers: *1. 36*

Lesson 6: Kp and partial pressures

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ calculate partial pressures and Kp

Lesson notes

KEY POINTS

Partial pressure p_A = mole fraction of A × total pressure; total pressure = sum of partial pressures. For gases: Kp = $(p_C)^c(p_D)^d/((p_A)^a(p_B)^b)$

Worked examples

Example: At equilibrium, a mixture at 2.00 atm contains 0.40 mol N_2O_4 and 0.60 mol NO_2. Find Kp for $N_2O_4 \rightleftharpoons 2NO_2$.

Mole fractions: N_2O_4 0.40, NO_2 0.60

$p(N_2O_4) = 0.80$ atm; $p(NO_2) = 1.20$ atm

$Kp = 1.20^2/0.80 = 1.8$ atm

Example: N_2O_4 is 20% dissociated at 1.00 atm. Find Kp.

Start 1 mol: N_2O_4 0.8, NO_2 0.4; total 1.2 mol

$p(N_2O_4) = 0.8/1.2 = 0.667$ atm; $p(NO_2) = 0.4/1.2 = 0.333$ atm

$Kp = 0.333^2/0.667 = 0.167$ atm

Practice Questions: Lesson 6

1. At equilibrium at 2.0 atm, a mixture contains 0.50 mol N_2O_4 and 0.50 mol NO_2. Find Kp for $N_2O_4 \rightleftharpoons 2NO_2$.

Answers: *1. 1.0 atm (each partial pressure = 1.0 atm)*

Lesson 7: Significance of K and factors affecting it

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ interpret the size of K and use Q to predict direction

Lesson notes

- **Large K (≫ 1):** equilibrium lies far to the right (reaction nearly complete). **Small K (≪ 1):** lies to the left (little product).
- K gives no information about the **rate** of reaching equilibrium.
- **Only temperature changes K.** Concentration, pressure and catalysts change the position or rate but not K.
- For an exothermic forward reaction, K **decreases** as temperature rises; for an endothermic forward reaction, K **increases**.
- **Reaction quotient Q:** the same expression using non-equilibrium concentrations. If Q < K, the reaction moves right; if Q > K, it moves left.

EXAM TIP

Reversing an equation gives K' = 1/K; doubling the coefficients gives K' = K^2.

Practice Questions: Lesson 7

1. Kc = 4.0 and Q = 1.5. In which direction will the reaction move?

Answers: *1. to the right (forward), because Q < Kc*

Lesson 8: Heterogeneous equilibria

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ write K for heterogeneous equilibria

Lesson notes

KEY POINTS

In heterogeneous equilibria, the concentrations of **pure solids and pure liquids** are constant and are left out of K

Worked examples

Example: Write Kc or Kp for (a) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ (b) $Fe_3O_4(s) + 4H_2(g) \rightleftharpoons 3Fe(s) + 4H_2O(g)$.

(a) $K_p = p(CO_2)$ (the pressure of CO_2 is fixed at a given temperature)

(b) $K_c = [H_2O]^4/[H_2]^4$

Example: For $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = 0.23$ atm at 1000 K. What happens if the CO_2 pressure is kept at 0.10 atm?

$Q = 0.10 < K_p$: $CaCO_3$ continues to decompose (CO_2 is removed as it forms: this is why lime kilns are open)

Practice Questions: Lesson 8

1. Write Kp for $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$.

Answers: *1. $K_p = p(CO_2)$*

Lesson 9: Temperature, enthalpy change and K

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ relate the change of K with temperature to ΔH

Lesson notes

KEY POINTS

van 't Hoff relationship: $\ln K = -\Delta H/(RT) + \text{constant}$; $\ln(K_2/K_1) = (\Delta H/R)(1/T_1 - 1/T_2)$

- A plot of ln K against 1/T is a straight line of gradient −ΔH/R.
- Exothermic (ΔH < 0): K falls as T rises. Endothermic (ΔH > 0): K rises as T rises.

Worked examples

Example: For a reaction, $K = 1.0 \times 10^{-3}$ at 300 K and 4.0×10^{-3} at 350 K. Find ΔH.

$\ln 4 = (\Delta H/8.31)(1/300 - 1/350) = (\Delta H/8.31)(4.76 \times 10^{-4})$

$\Delta H = 1.386 \times 8.31/(4.76 \times 10^{-4}) = 2.42 \times 10^{4}$ J mol^{-1} = +24.2 kJ mol^{-1} (endothermic)

Practice Questions: Lesson 9

1. K decreases as temperature rises. Is the forward reaction exothermic or endothermic?

Answers: *1. exothermic*

Lesson 10: Redox systems: half-cells and cells

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define the terms used in electrochemistry

Lesson notes

KEY POINTS

Oxidation = loss of electrons (increase in oxidation number); **reduction** = gain of electrons (decrease). OIL RIG

- **Half-cell:** an element in contact with a solution of its ions (or two ions of the same element), e.g. Zn(s) in Zn^{2+}(aq).
- **Electrochemical (galvanic) cell:** two half-cells joined by a salt bridge and an external circuit, converting chemical energy into electrical energy.
- **Anode:** electrode where oxidation occurs (negative in a galvanic cell). **Cathode:** where reduction occurs (positive).
- **Redox (electrode) potential:** the potential difference set up between a half-cell and the standard hydrogen electrode; a measure of the tendency of the half-cell to be reduced.

Practice Questions: Lesson 10

1. At which electrode does oxidation occur in a galvanic cell, and what is its sign?

Answers: *1. the anode; negative*

Lesson 11: Redox equilibria and balancing redox equations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ balance redox equations

Lesson notes

- At each electrode a dynamic equilibrium exists: $Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$. Electrode potentials are written as **reductions**.
- **Balancing:** write the half-equations, balance atoms (O with H_2O, H with H^+), balance charge with electrons, multiply so the electrons cancel, add.

Worked examples

Example: Balance the reaction between MnO_4^- and SO_2 in water ($SO_2 \rightarrow SO_4^{2-}$).

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (× 2)

$SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e^-$ (× 5)

$2MnO_4^- + 5SO_2 + 2H_2O \rightarrow 2Mn^{2+} + 5SO_4^{2-} + 4H^+$

Practice Questions: Lesson 11

1. Balance: $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$ (acid).

Answers: *1. $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$*

Lesson 12: Types of half-cells

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the types of half-cell

Lesson notes

Type	Example	Electrode
Metal / metal ion	$Cu^{2+}(aq)/Cu(s)$	the metal itself
Non-metal / non-metal ion	$Cl_2(g)/Cl^-(aq)$; $H^+(aq)/H_2(g)$	platinum (inert), gas bubbled over it
Ion / ion (same element, two oxidation states)	$Fe^{3+}(aq)$, $Fe^{2+}(aq)$; MnO_4^-, Mn^{2+} in H^+	platinum dipped in a solution containing both ions (1 mol dm^{-3} each)

- Platinum is used because it is inert and conducts electrons.

Practice Questions: Lesson 12

1. Describe the half-cell for the Fe^{3+}/Fe^{2+} system.

Answers: *1. a platinum electrode in a solution containing 1 mol dm^{-3} Fe^{3+} and 1 mol dm^{-3} Fe^{2+}*

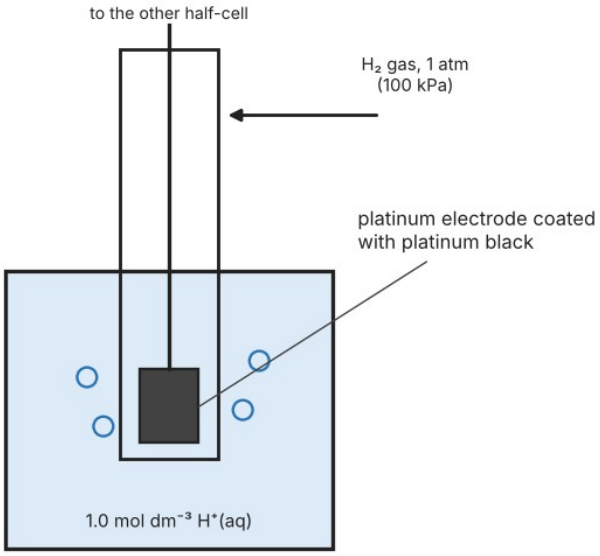
Lesson 13: Standard electrode potentials and the SHE

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe the SHE and define standard electrode potential

Lesson notes



298 K; E° defined as 0.00 V

KEY POINTS

Standard electrode potential E°: the e.m.f. of a cell formed by a half-cell connected to the standard hydrogen electrode, under standard conditions: 298 K, 1 mol dm^{-3} solutions, 100 kPa (1 atm) gases, Pt where needed. $E°(H^+/H_2) = 0.00$ V

Half-equation	E° / V
$Li^+ + e^- \rightleftharpoons Li$	−3.04
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	−0.44
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+0.34
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.54
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77
$Ag^+ + e^- \rightleftharpoons Ag$	+0.80
$Br_2 + 2e^- \rightleftharpoons 2Br^-$	+1.07
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	+1.51
$F_2 + 2e^- \rightleftharpoons 2F^-$	+2.87

- More **negative** E°: the reduced form is a better reducing agent (Zn, Li). More **positive** E°: the oxidised form is a better oxidising agent (F_2, MnO_4^-).

Practice Questions: Lesson 13

1. State the standard conditions for measuring E°.

Answers: *1. 298 K; 1 mol dm^{-3} solutions; 100 kPa gas pressure; platinum where needed*

END OF FREE SAMPLE

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